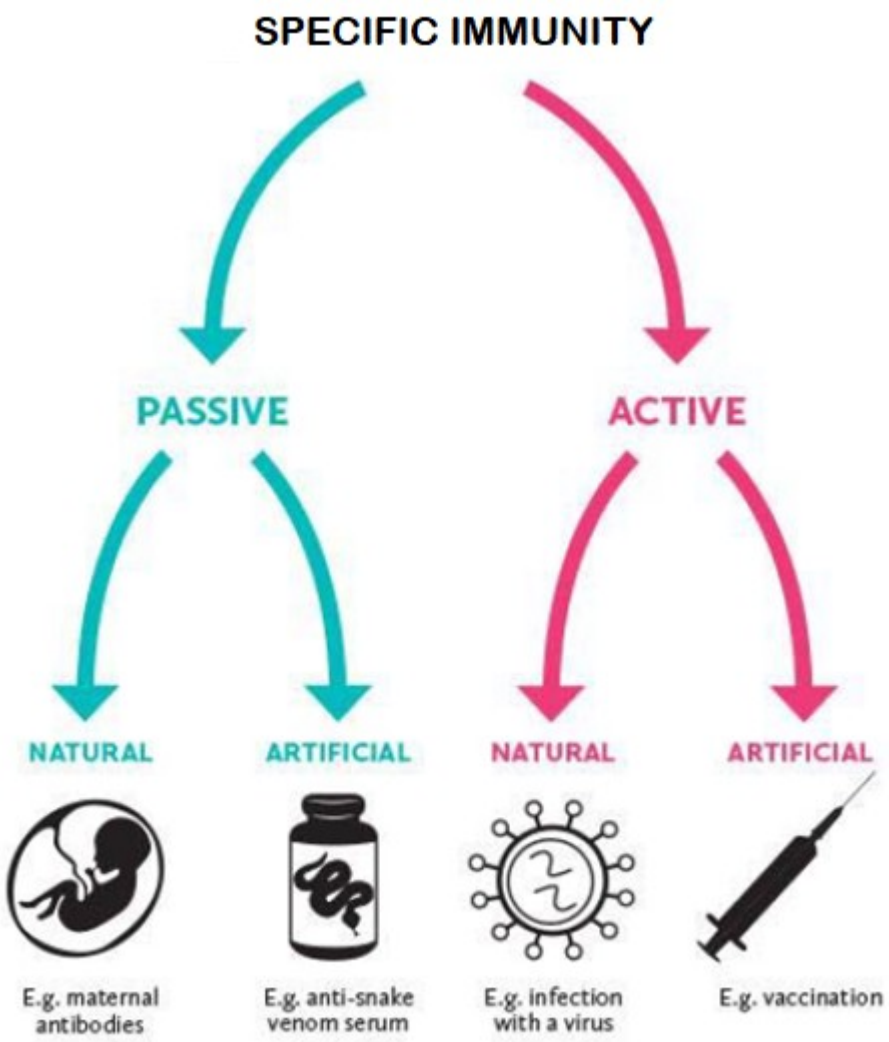
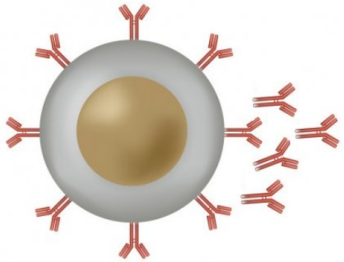
Types of Specific Immunity



Active vs Passive Immunity



ACTIVE

- Exposure to antigen causes body to *actively* produce immunity
  - sensitisation of lymphocytes, producing cell mediated and antibody mediated immunity.
  - memory cells.

*Eg: When a virus naturally enters your body, your body **actively** fights it off via the process above.*



*Eg: When you get a vaccine, your body **actively** produces antibody via the process above*





PASSIVE

- Body gets "ready made" antibody from another source – body does not need to actively work to produce the antibody. Body is not exposed to the antigen, so active immunity can't develop.
  - Antibody received neutralises any pathogens present but will be used up.
  - No memory cells produced

Eg: Antibody produced by the mother passes through the placenta and breast milk to the baby



Eg: Snake antivenom is bottled antibody that is injected into the blood to neutralise ven



Learning Aim: Explain the differences between active and passive immunity

Natural vs Artificial Immunity

NATURAL

- Immunity develops via natural processes in the human body.

Eg: When a virus naturally enters your body, your body actively fights it off via cell-mediated and anti-body mediated immunity.



Eg: Antibody naturally produced by the body are passed from mother to baby via the placenta and breast milk so that the baby becomes temporarily, passively immune.



ARTIFICIAL

- Scientific processes are used to create immunity.

Eg: vaccines are developed to produce active immunity without illness.

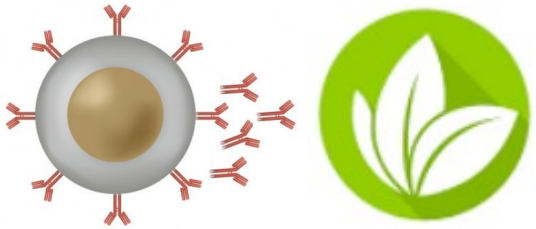


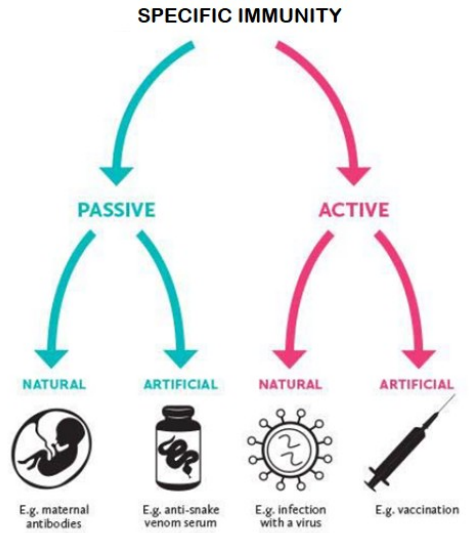
Eg: Snake antivenom is produced so that antibodies to the venom can be injected into a person bitten by a snake to neutralise the venom.



Learning Aim: Explain the differences between natural and artificial immunity

Example: Active Natural Immunity





- Eg: Getting the Measles
- Measles is a virus that occurs ***naturally***.
- If someone is exposed to the Measles virus, their immune system responds ***actively***.
  - B-cells and T-cells (lymphocytes) become sensitised
  - produce antibody to fight the virus via *Antibody Mediated (Humoral) Immunity*
  - Produce memory cells specific to the virus so next encounter will be neutralised before symptoms develop

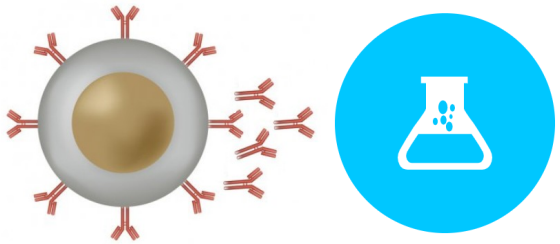


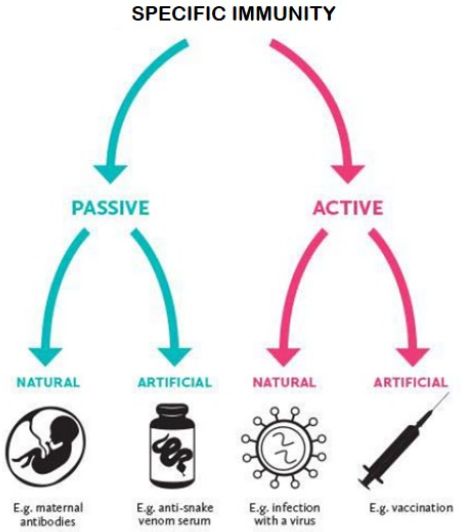




Learning Aim: Explain how active natural immunity occurs, with an example

Example: Active Artificial Immunity





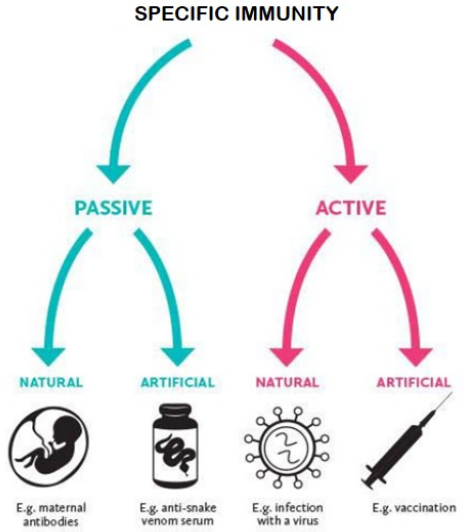
- EG: Receiving a Measles Vaccine
- The measles vaccine is ***artificial – it is man-made***
- A vaccine is given, containing either killed measles virus, substances that mimic the measles antigen, or a very mild form of the virus. The body then responds ***actively*** to produce long lasting immunity.
  - B-cells become sensitised
  - Antibody produced via antibody-mediated immunity
  - Memory cells produced
  - Immunity to measles results.



Learning Aim: Explain how active artificial immunity occurs, with an example

Example: Passive Natural Immunity





- EG: child gets antibodies via breastfeeding or across placenta
- The antibodies are made ***naturally*** by the mother's immune system.
- Baby gets antibodies from breastmilk or across placenta. The baby gets the antibody without the baby's immune system having to do anything. The process is ***passive.***

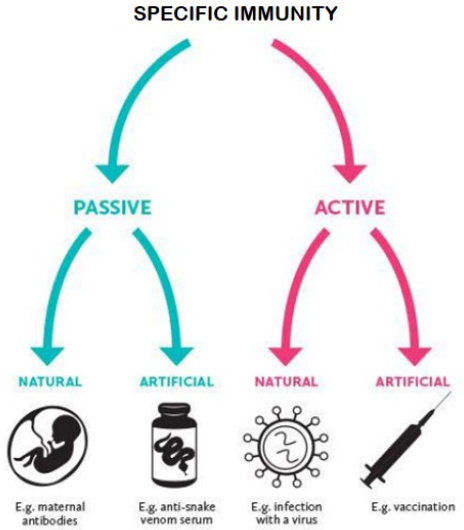
  - Baby then has circulating antibody to neutralise specific pathogen.
  - Doesn't last – no active response from baby's immune system, no memory-cells produced.



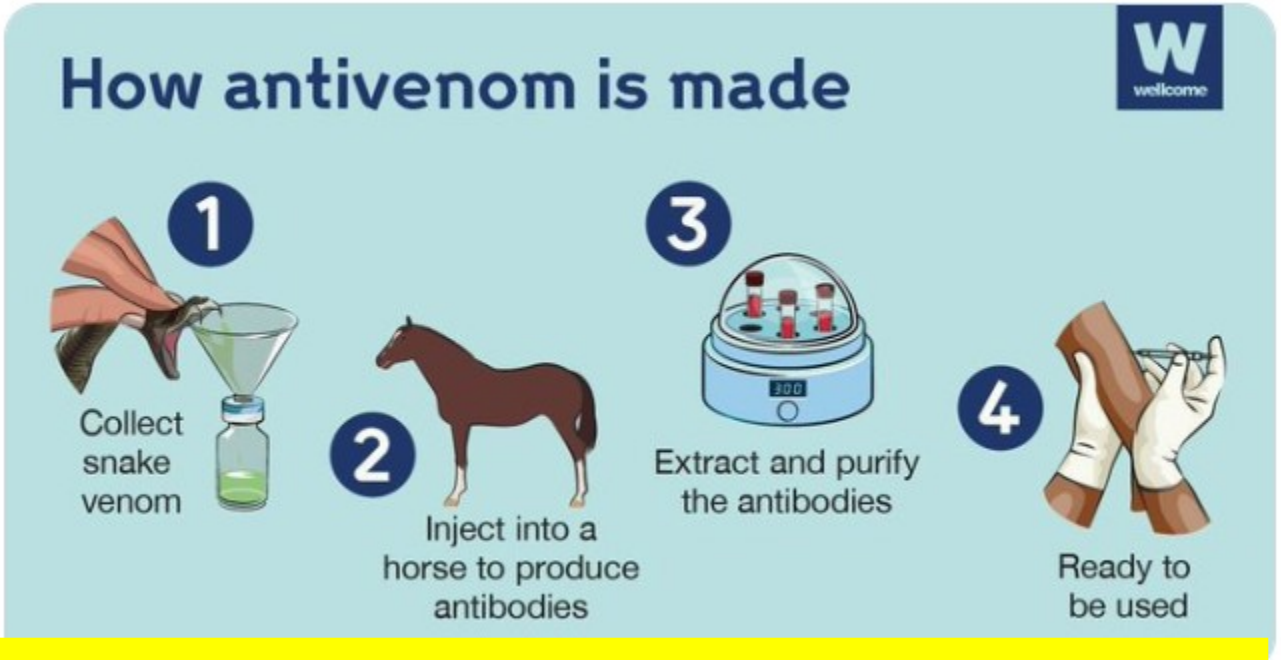
Learning Aim: Explain how passive natural immunity occurs, with an example

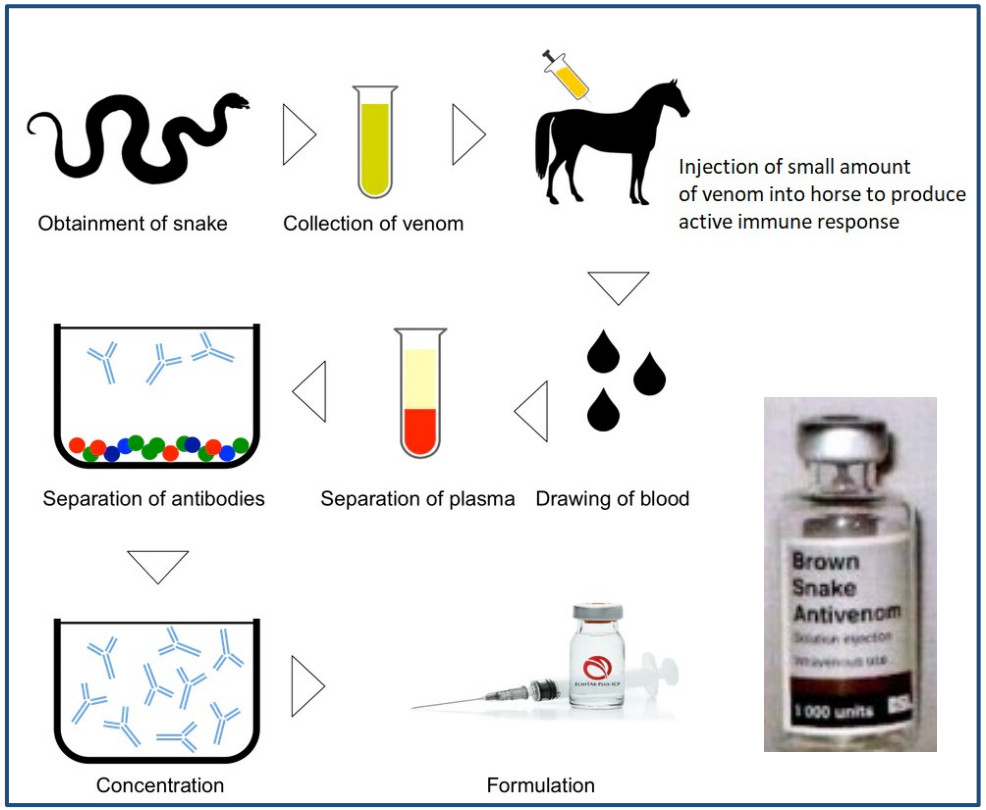
Example: Passive Artificial Immunity





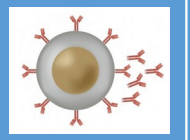
- Eg: administration of snakebite antivenom
- Antibodies to snake venom produced via a man-made process ***(artificial)***
- Antibody injected into patient's bloodstream:
  - Antibody neutralises venom
  - Body does not work to develop immunity – gets antibody ***passively***
  - Short term only – no memory-cells produced





Learning Aim: Explain how passive artificial immunity occurs, with an example

Summary Table: Types of Immunity

	Active Immunity 	Passive Immunity 
	• Body actively produces antibody • Antibody mediated process • Memory cells produced • Long term immunity – more antibody can be produced each time pathogen is encountered.	• Antibodies are gained from another source • Body gains antibody passively – immune system not involved. • No memory cells produced • Immunity only lasts until antibody used up.
Natural 	Catching tetanus • Tetanus toxin enters body, B-cells activated, antibody produced actively. • Immune system will recognise toxin next time it is encountered.	Getting tetanus antibodies from mothers' milk • Antibodies will neutralise toxin if present but immune system isn't activated, so no future immunity.
Artificial 	Getting the tetanus Vaccine • Tetanus toxoid vaccine enters body, B-cells activated, antibody produced actively. • Immune system will recognise toxin next time it is encountered.	Getting man-made tetanus antitoxin injected • Antibodies in antitoxin will neutralise toxin if present but immune system isn't activated, so no future immunity.